Technical Service Bulletin

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Engine No Start Issue	

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Core Issue

This Technical Service Bulletin alerts the field to an issue with QST30 engines being built with the wrong relay in the engine lift pump harness. Engines were built with a 12 volt relay, Part Number 3658948, rather than the specified 24 volt relay, Part Number 3658780. This causes the 12 volt relay to become overloaded and fail. This prevents power from being supplied to the engine lift pumps and causes a no start condition at the next attempt to start the engine.

Confirmation

This issue affects all QST30 engines. The suspected build range is estimated to be October 2009 to March 2010.

This issue is observed as a no start condition. The 12 volt relay becomes overloaded and typically fails within the first few running hours of a new engine. Depending on the fuel tank set up, the engine can remain running after the relay has malfunctioned. However, at the next attempt to start the engine, the no start condition will appear.

A 12 volt relay installed on the engine will show a lower resistance than the 24 volt relay across pins 85 and 86 on the lift pump harness connector. The resistance across pins 85 and 86 with the proper 24 volt relay installed will be in the range of 360-380 ohms. The resistance with the 12 volt relay installed will be approximately 75 ohms.

The supplier of the lift pump wiring harness installed the incorrect relay in the harness.

Overheated relay. No progressive damage is expected.

Resolution

If a 12 volt relay is found damaged, or a low resistance is found across pins 85 and 86 of the lift pump harness, replace the relay with the 24 volt relay, Part Number 3658780.

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

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